

Question	Answer	Mark	Mark scheme	Additional guidance
19	Proof	P1	for start to process to find area of $ABCDEF$, eg area of equilateral triangle $= \frac{1}{2} \times x \times x \times \sin 60 (= \frac{\sqrt{3}}{4}x^2)$ OR for start to process to find area of $FGHIJK$, eg area of equilateral triangle $= \frac{1}{2} \times px \times px \times \sin 60 (= \frac{\sqrt{3}}{4}p^2x^2)$	Any correct process to find the area of part of the hexagon is acceptable for this mark, eg $\frac{1}{2} \times x \times x \times \sin 120$ or $\frac{1}{2} \times (x + 2x) \times \frac{\sqrt{3}}{2}x$
		P1	for complete process of finding area of $ABCDEF$, eg $6 \times \frac{1}{2} \times x \times x \times \sin 60$ or $6 \times \frac{1}{2} \times x \times x \times \frac{\sqrt{3}}{2} \left(= \frac{3\sqrt{3}}{2}x^2 \right)$ oe OR for complete process of finding area of $FGHIJK$, eg $6 \times \frac{1}{2} \times px \times px \times \frac{\sqrt{3}}{2} \left(= \frac{3\sqrt{3}}{2}p^2x^2 \right)$ oe	Allow $\sin 60$ left in expressions for the first 3 marks.
		P1	for process of finding area of $ABCDEF$ eg $\frac{3\sqrt{3}}{2}x^2$ oe AND for process of finding area of $FGHIJK$, eg $p^2 \times \frac{3\sqrt{3}}{2}x^2$ oe	
		C1	correct algebra leading to given result, $\frac{3\sqrt{3}}{2}(p^2 - 1)x^2$	Accept $\frac{3\sqrt{3}}{2}x^2(p^2 - 1)$ as final result.